

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $X = \{2, 4, 6, 8\}$, $Y = \{2, 3, 4, 5, 6\}$, and $Z = \{1, 2, 3, 8, 9\}$. List the members of each set, using set braces.

14. $X \cup Y$

15. $X \cap Y$

16. X'

17. $X' \cap Y'$

18. $X' \cap (Y' \cup Z)$

19. $(X \cap Y) \cup (X \cap Z)$

20. $(X \cap Y) \cap \{1, 7\}$

Answer the following questions.

21. There are 1500 students at JCSU. If 100 are math majors, 800 are female, and 75 are female math majors.

a. How many students are not math majors?

b. How many students are both math majors and female?

c. How many students are not math majors or female?

22. Of the students in your class, 11 have sisters, 14 have brothers, 10 have a brother and a sister, and 5 have neither a brother or a sister.

a. How many students have a brother or a sister?

b. How many students are in the class?